Haan! NIELIT O Level M4 (IoT) mein in practicals ke aane ke chances kaafi zyada hote hain. Inhe zaroor practice karna.

---

1. Single LED ON

```
int led = 13;

void setup()
{
  pinMode(led, OUTPUT);
}

void loop()
{
  digitalWrite(led, HIGH);
}
```

---

2. LED Blink

```
int led = 13;

void setup()
{
  pinMode(led, OUTPUT);
}
```

```
void loop()
{
  digitalWrite(led, HIGH);
  delay(1000);

  digitalWrite(led, LOW);
  delay(1000);
}
```

---

3. Push Button + LED

```
int led = 13;
int button = 2;

void setup()
{
  pinMode(led, OUTPUT);
  pinMode(button, INPUT);
}

void loop()
{
  if(digitalRead(button)==HIGH)
    digitalWrite(led,HIGH);
  else
    digitalWrite(led,LOW);
}
```

---

4. Buzzer

```
int buzzer = 8;

void setup()
{
  pinMode(buzzer, OUTPUT);
}

void loop()
{
  digitalWrite(buzzer,HIGH);
```

```
  delay(1000);

  digitalWrite(buzzer,LOW);
  delay(1000);
}
```

---

5. LED + Buzzer

```
int led=13;
int buzzer=8;

void setup()
{
  pinMode(led,OUTPUT);
  pinMode(buzzer,OUTPUT);
}

void loop()
{
  digitalWrite(led,HIGH);
  digitalWrite(buzzer,HIGH);
  delay(1000);

  digitalWrite(led,LOW);
  digitalWrite(buzzer,LOW);
  delay(1000);
}
```

---

6. RGB LED

```
int red=5;
int green=6;
int blue=7;

void setup()
{
  pinMode(red,OUTPUT);
  pinMode(green,OUTPUT);
  pinMode(blue,OUTPUT);
}
```

```
void loop()
{
  digitalWrite(red,HIGH);
  digitalWrite(green,LOW);
  digitalWrite(blue,LOW);
  delay(1000);

  digitalWrite(red,LOW);
  digitalWrite(green,HIGH);
  digitalWrite(blue,LOW);
  delay(1000);

  digitalWrite(red,LOW);
  digitalWrite(green,LOW);
  digitalWrite(blue,HIGH);
  delay(1000);
}
```

---

7. LDR (Light Sensor)

```
int ldr=A0;

void setup()
{
  Serial.begin(9600);
}

void loop()
{
  int value=analogRead(ldr);
  Serial.println(value);
  delay(500);
}
```

---

8. LDR + LED

```
int led=13;
int ldr=A0;

void setup()
{
```

```
  pinMode(led,OUTPUT);
}

void loop()
{
  int value=analogRead(ldr);

  if(value<500)
    digitalWrite(led,HIGH);
  else
    digitalWrite(led,LOW);

  delay(200);
}
```

---

9. Ultrasonic Sensor

```
int trig=9;
int echo=10;

void setup()
{
  pinMode(trig,OUTPUT);
  pinMode(echo,INPUT);
  Serial.begin(9600);
}

void loop()
{
  digitalWrite(trig,LOW);
  delayMicroseconds(2);

  digitalWrite(trig,HIGH);
  delayMicroseconds(10);

  digitalWrite(trig,LOW);

  long duration=pulseIn(echo,HIGH);
  int distance=duration*0.034/2;

  Serial.println(distance);
  delay(500);
}
```

---

10. Temperature & Humidity (DHT11)

```
#include <DHT.h>

#define DHTPIN 2
#define DHTTYPE DHT11

DHT dht(DHTPIN,DHTTYPE);

void setup()
{
  Serial.begin(9600);
  dht.begin();
}

void loop()
{
  Serial.print("Temp = ");
  Serial.println(dht.readTemperature());

  Serial.print("Humidity = ");
  Serial.println(dht.readHumidity());

  delay(2000);
}
```

---

11. Servo Motor

```
#include <Servo.h>

Servo myservo;

void setup()
{
  myservo.attach(9);
}

void loop()
{
  myservo.write(0);
  delay(1000);

  myservo.write(90);
  delay(1000);
```

```
  myservo.write(180);
  delay(1000);
}
```

⭐ Most Important Viva Questions

1. `pinMode()` ka use?
2. `digitalWrite()` kya karta hai?
3. `digitalRead()` kya karta hai?
4. `analogRead()` aur `digitalRead()` mein difference?
5. `delay()` aur `delayMicroseconds()` mein difference?
6. `pulseIn()` kis sensor ke saath use hota hai?
7. `Serial.begin(9600)` kyon likhte hain?
8. `HIGH` aur `LOW` ka matlab?
9. `setup()` kitni baar chalta hai?
10. `loop()` kitni baar chalta hai?
11. RGB LED mein kitni pins hoti hain?
12. DHT11 kya measure karta hai?
13. LDR kis cheez ko detect karta hai?
14. Ultrasonic sensor kisliye use hota hai?
15. Servo motor kitne degree tak rotate karta hai?

Ye **15 practicals + 15 viva questions** exam ke liye sabse important hain. Inki practice kar loge to NIELIT O Level M4 practical ke liye achchi taiyari ho jayegi.

in sab ka common code doo and yea batao ki ky change karna hai

Haan! Ye sabse important trick hai. Agar **common template** yaad ho gaya to tum **80–90% Arduino practicals** khud bana loge.

Arduino Common Template

```
// Variable Declaration

void setup()
{
   // pinMode
}

void loop()
{
   // Main Code
}
```

Ya phir:

```
int device = 13;

void setup()
{
   pinMode(device, OUTPUT);
}

void loop()
{

}
```

---

1. LED

Sirf ye change hoga:

```
int led = 13;

pinMode(led, OUTPUT);

digitalWrite(led, HIGH);
digitalWrite(led, LOW);
```

---

2. Buzzer

Sirf naam change

```
int buzzer = 8;

pinMode(buzzer, OUTPUT);

digitalWrite(buzzer, HIGH);
digitalWrite(buzzer, LOW);
```

Bas LED ki jagah **buzzer**.

---

3. Button

Yahan OUTPUT nahi INPUT hoga.

```
int button = 2;

pinMode(button, INPUT);

digitalRead(button);
```

Difference:

```
LED/Buzzer → OUTPUT + digitalWrite()

Button → INPUT + digitalRead()
```

---

4. RGB

Ek LED ki jagah 3 LED

```
int red=5;
int green=6;
int blue=7;
```

PinMode

```
pinMode(red,OUTPUT);
pinMode(green,OUTPUT);
pinMode(blue,OUTPUT);
```

ON/OFF

```
digitalWrite(red,HIGH);
digitalWrite(green,LOW);
digitalWrite(blue,LOW);
```

---

5. Ultrasonic

Variables

```
int trig=9;
int echo=10;
```

PinMode

```
pinMode(trig,OUTPUT);
pinMode(echo,INPUT);
```

Extra

```
C++
```

---

6. DHT11

Sirf library add hoti hai

```
C++
```

Aur

```
dht.begin();

dht.readTemperature();

dht.readHumidity();
```

---

7. Servo

Library

```
C++
```

Object

```
C++
```

Setup

```
C++
```

Loop

```
C++
```

---

8. LDR

Variable

```
C++
```

Reading

```
C++
```

Sabse Important Pattern

OUTPUT Device

```
Variable

↓

pinMode(...,OUTPUT)

↓

digitalWrite(HIGH/LOW)
```

Examples

- LED
- Buzzer
- Relay

INPUT Device

```
Variable

↓

pinMode(...,INPUT)

↓

digitalRead()
```

Examples

- Button
- Switch

ANALOG INPUT

```
Variable

↓

analogRead()
```

Examples

- LDR
- Potentiometer

SERIAL OUTPUT

```
Serial.begin(9600);

↓

Serial.print()

↓

Serial.println()
```

Examples

- DHT11
- Ultrasonic
- LDR

🔥 5-Line Trick (Exam ke liye)

LED/Buzzer

```
pinMode(...,OUTPUT);
digitalWrite(...,HIGH);
digitalWrite(...,LOW);
```

Button

```
pinMode(...,INPUT);
```

C++

Sensor

C++

Distance

C++

Serial Monitor

```
Serial.begin(9600);
```

📌 Yaad rakhne ki Short Trick

Device	PinMode	Function
LED	OUTPUT	`digitalWrite()`
Buzzer	OUTPUT	`digitalWrite()`
RGB	OUTPUT	`digitalWrite()`
Button	INPUT	`digitalRead()`
LDR	INPUT (Analog)	`analogRead()`
Ultrasonic	Trig = OUTPUT, Echo = INPUT	`pulseIn()`
DHT11	Library	`readTemperature()` , `readHumidity()`
Servo	Library	`attach()` , `write()`

Agar tumhare **kal practical** hai, to main **1-page "Arduino Cheat Sheet"** bhi bana sakta hoon jisme sirf wahi lines hongi jo har practical mein change hoti hain. Exam se 5 minute

pehle dekhoge to lagbhag saare basic practical yaad aa jayenge.

ldr, ultrasonic ,dht 11 , servo ka code doo

Bilkul. Ye **NIELIT O Level M4 Practical** ke exam-oriented aur simple codes hain.

---

1. LDR (Light Dependent Resistor)

Code

```
int ldr = A0;

void setup()
{
  Serial.begin(9600);
}

void loop()
{
  int value = analogRead(ldr);

  Serial.println(value);

  delay(500);
}
```

Yaad rakhne ki Trick

- LDR → `A0`
- `analogRead()`
- `Serial.println()`

---

2. Ultrasonic Sensor (HC-SR04)

Code

```
int trig = 9;
int echo = 10;

long duration;
int distance;

void setup()
{
  pinMode(trig, OUTPUT);
  pinMode(echo, INPUT);

  Serial.begin(9600);
}

void loop()
{
  digitalWrite(trig, LOW);
  delayMicroseconds(2);

  digitalWrite(trig, HIGH);
  delayMicroseconds(10);

  digitalWrite(trig, LOW);

  duration = pulseIn(echo, HIGH);

  distance = duration * 0.034 / 2;

  Serial.println(distance);

  delay(500);
}
```

Yaad rakhne ki Trick

- Trig → OUTPUT
- Echo → INPUT
- `pulseIn()`
- `distance = duration * 0.034 / 2`

---

3. DHT11 (Temperature & Humidity)

Code

```
#include <DHT.h>

#define DHTPIN 2
#define DHTTYPE DHT11

DHT dht(DHTPIN, DHTTYPE);

void setup()
{
  Serial.begin(9600);
  dht.begin();
}

void loop()
{
  float temp = dht.readTemperature();
  float hum = dht.readHumidity();

  Serial.print("Temperature = ");
  Serial.println(temp);

  Serial.print("Humidity = ");
  Serial.println(hum);

  delay(2000);
}
```

Yaad rakhne ki Trick

- `#include <DHT.h>`
- `dht.begin()`
- `readTemperature()`
- `readHumidity()`

---

4. Servo Motor

Code

```
#include <Servo.h>

Servo myservo;

void setup()
```

```
{
  myservo.attach(9);
}

void loop()
{
  myservo.write(0);
  delay(1000);

  myservo.write(90);
  delay(1000);

  myservo.write(180);
  delay(1000);
}
```

Yaad rakhne ki Trick

- `#include <Servo.h>`
- `Servo myservo;`
- `attach(9)`
- `write(angle)`

📌 Ek Nazar Mein (Cheat Sheet)

Device	Important Function
LDR	`analogRead(A0)`
Ultrasonic	`pulseIn()`
DHT11	`readTemperature()` , `readHumidity()`
Servo	`attach()` , `write()`

Viva Questions

LDR

- Q. LDR kya measure karta hai?
- A. Light intensity.

Ultrasonic

- Q. `pulseIn()` ka use kya hai?
- A. Echo pulse ki duration measure karne ke liye.

DHT11

- Q. DHT11 kya measure karta hai?
- A. Temperature aur Humidity.

Servo

- Q. Servo motor kitne degree tak rotate karti hai?
- A. Aam taur par **0° se 180°** tak.

Ye codes NIELIT O Level practical ke liye standard aur exam-friendly hain.

ChatGPT can make mistakes. Check important info.